MaintenAI – Predictive Maintenance Reference Guide

Educational RAG knowledge document based on the UCI AI4I 2020 Predictive Maintenance Dataset

Purpose and scope

This document provides a compact reference for MaintenAI's retrieval-augmented generation (RAG) demonstration. It describes the five failure-mode labels used by the AI4I 2020 dataset and gives cautious, general follow-up guidance. It is an educational reference, not an original equipment manufacturer's (OEM) service manual and not a substitute for machine-specific instructions, qualified technicians, or workplace safety procedures.

1. Sensor measurements used by the dataset

Measurement	Unit	What it represents
Air temperature	K	Ambient/air temperature around the simulated process.
Process temperature	K	Temperature associated with the simulated process.
Rotational speed	rpm	Machine/tool rotational speed.
Torque	Nm	Rotational force/load measurement.
Tool wear	min	Accumulated tool-use/wear measure in the dataset.

2. Failure modes described by AI4I 2020

TWF – Tool Wear Failure

The dataset models tool wear failure when the tool reaches a randomly selected wear interval between 200 and 240 minutes. This is a dataset rule, not a universal replacement interval.

HDF – Heat Dissipation Failure

The dataset's condition combines a small air-to-process temperature difference (below 8.6 K) with rotational speed below 1,380 rpm. These thresholds belong to the synthetic dataset specification.

PWF – Power Failure

The dataset models failure when calculated process power is below 3,500 W or above 9,000 W. The calculation uses torque and rotational speed.

OSF – Overstrain Failure

The dataset models overstrain when tool wear multiplied by torque exceeds a product-type-specific threshold: 11,000 minNm for L, 12,000 minNm for M, and 13,000 minNm for H.

RNF – Random Failure

The dataset includes a small random-failure probability that is not explained by the measured process variables.

3. General troubleshooting workflow

1. Review the machine alarm, operating history, and sensor readings. Compare readings with the equipment manufacturer's specified operating limits.

2. If an abnormal condition or safety alarm is present, follow the site's shutdown and isolation procedure. Do not bypass guards, interlocks, or protective devices.

3. Have an authorized technician inspect the relevant subsystem using the correct OEM service documentation and approved tools.

4. Record the observed condition, actions taken, replaced components, and post-maintenance test results.

5. Return equipment to service only after the responsible person confirms that required inspections and safety checks are complete.

4. How MaintenAI should use this reference

The RAG component should retrieve the relevant passages above and provide a concise explanation. It should identify when a question cannot be answered from this document, avoid inventing machine-specific limits, and direct the user to the OEM manual or a qualified technician for actual repair decisions. A machine-learning prediction is an estimate based on the model and input data; it is not proof that a machine is safe or faulty.

5. Source and attribution

Source: UCI Machine Learning Repository, “AI4I 2020 Predictive Maintenance Dataset” (2020), DOI: 10.24432/C5HS5C. The repository describes the dataset as synthetic and documents the five failure-mode rules. This MaintenAI reference guide paraphrases those dataset definitions and adds general educational safety guidance.

Source page: https://archive.ics.uci.edu/dataset/601/ai4i

Important limitation: The AI4I 2020 dataset simulates a milling-machine scenario. Its thresholds and labels must not be treated as universal industrial maintenance standards.